

Tristan Ryoma Fuchs

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RESEARCH INTERESTS

Superconductivity · first-principles (DFT) calculations. Extended Ginzburg–Landau framework for the coherence lengths and magnetic penetration depth of multiband superconductors, toward first-principles prediction of superconducting properties.

EDUCATION

- Apr 2025 – present **Master's Program, Department of Physics, Graduate School of Science, The University of Tokyo**
Supervisor: Prof. Ryotaro Arita. MERIT-WINGS program student since Oct 2025.
- Apr 2021 – Mar 2025 **B.Sc. in Physics, Department of Physics, Faculty of Science, Tohoku University**
Supervisor: Prof. Takashi Koretsune.
Undergraduate research: First-principles Migdal–Eliashberg calculation of the superconducting gap functions of Al and H₃S
- Sep 2016 – Jun 2020 **Shanghai Jincai High School, International Division, Shanghai, China**
Grades 9–12; International Baccalaureate Diploma Programme (Grades 11–12).
- Apr 2008 – Aug 2016 **Shanghai Japanese School, Pudong Campus, Shanghai, China**
Grades 1–9.

PUBLICATIONS

1. **T. R. Fuchs**, T. Nomoto, H. Watanabe, and R. Arita, “Microscopic calculation of coherence lengths and magnetic penetration depth in multiband superconductors,” [arXiv:2609.06014](https://arxiv.org/abs/2609.06014) [cond-mat.supr-con] (2026), under review.

PRESENTATIONS

International conferences and schools

- Sep 2026 **Microscopic Calculation of Characteristic Length Scales in Multi-band Superconductors**
POSTER **SCHEDULED**
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — International Conference on Strongly Correlated Electron Systems (SCES 2026), Toyama, Japan
- Sep 2026 **Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Multi-Band Superconductors** **POSTER** **SCHEDULED**
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — International Conference on Asymmetric Quantum Matters (Asymmetry 2026), Osaka, Japan
- Feb 2026 **Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Two Band Superconductors** **POSTER**
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — CEMS Topical Meeting on Recent Trends in Superconductivity, RIKEN, Wako, Japan
- Sep 2025 **Microscopic Calculation of Coherence Length and Magnetic Penetration Depth in Superconductors** **POSTER**
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — Autumn School on Correlated Electrons: Understanding Correlated Materials with DMFT, Forschungszentrum Jülich, Germany

Domestic conferences and schools (Japan)

- Sep 2026 **Calculation of Coherence Lengths and the Magnetic Penetration Depth Based on the Real-Space Bogoliubov–de Gennes Method** **POSTER** **SCHEDULED**
T. R. Fuchs, T. Nomoto, H. Watanabe, R. Arita — JPS 81st Annual Meeting (Physical Society of Japan), The University of Tokyo, Komaba (in Japanese)

Sep 2026	Microscopic Calculation of Characteristic Length Scales in Multi-band Superconductors Based on the Extended Ginzburg–Landau Method POSTER BEST POSTER AWARD <u>T. R. Fuchs</u> , T. Nomoto, H. Watanabe, R. Arita — MERIT Camp 2026 (MERIT-WINGS, The University of Tokyo), Minano, Saitama
Aug 2026	Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Multi-Band Superconductors Based on Extended Ginzburg–Landau Theory ORAL POSTER <u>T. R. Fuchs</u> , T. Nomoto, H. Watanabe, R. Arita — 71st Condensed Matter Physics Summer School (物性若手夏の学校), Aso, Kumamoto (in Japanese)
May 2026	Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Multi-Band Superconductors POSTER <u>T. R. Fuchs</u> , T. Nomoto, H. Watanabe, R. Arita — Kickoff Meeting of the Transformative Research Area (A) “Asymmetric Quantum Matters,” Nasu, Tochigi (in Japanese)
Mar 2026	Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Two-Band Superconductors ORAL <u>T. R. Fuchs</u> , T. Nomoto, H. Watanabe, R. Arita — JPS 2026 Spring Meeting (Physical Society of Japan), online (in Japanese)
Mar 2026	Microscopic Calculation of Coherence Lengths and Magnetic Penetration Depth in Two-Band Superconductors POSTER BEST POSTER AWARD <u>T. R. Fuchs</u> , T. Nomoto, H. Watanabe, R. Arita — Computational Physics Spring School 2026 (計算物理春の学校), Okinawa (in Japanese)

AWARDS AND FELLOWSHIPS

Sep 2026	Best Poster Award, MERIT Camp 2026 — MERIT-WINGS program, The University of Tokyo
Mar 2026	Best Poster Award, Computational Physics Spring School 2026 — 計算物理春の学校
Oct 2025 – present	MERIT-WINGS program student World-leading Innovative Graduate Study Program for Materials Research, Information, and Technology (MERIT-WINGS), The University of Tokyo

EXPERIENCE

Dec 2023 – present	Long-Term Intern (Student Staff Lead), Nagase Brothers Inc. Oct 2024 – present: development of internal automation tools at Yotsuya Otsuka Dec 2023 – Mar 2026: project management at Toshin Digital University
Spring 2024	Teaching Assistant, 学問論 (first-year general education course), Tohoku University

LEADERSHIP AND ACTIVITIES

Oct 2025 – Sep 2026	Deputy Organizer, MERIT Camp — The University of Tokyo
Oct 2023 – Mar 2025	Student Staff, Kita-Aobayama Library (Science & Pharmacy Library) — Tohoku University
Apr 2022 – Mar 2023	Vice President, 物理の会 (“Butsuri no Kai,” physics seminar society) — Tohoku University

CERTIFICATIONS AND TEST SCORES

2026	EIKEN Grade 1 実用英語技能検定 1級
2025	Applied Information Technology Engineer Examination 応用情報技術者試験 (IPA, Japan) — passed
2024	TOEFL iBT 105
2024	TOEIC Listening & Reading 960
2022	HSK Level 4 汉语水平考试 (Chinese Proficiency Test)

LANGUAGES AND SKILLS

Languages: Japanese (native), English (fluent), German (conversational), Chinese (basic)
Programming: Julia (main language for research), Python, Google Apps Script